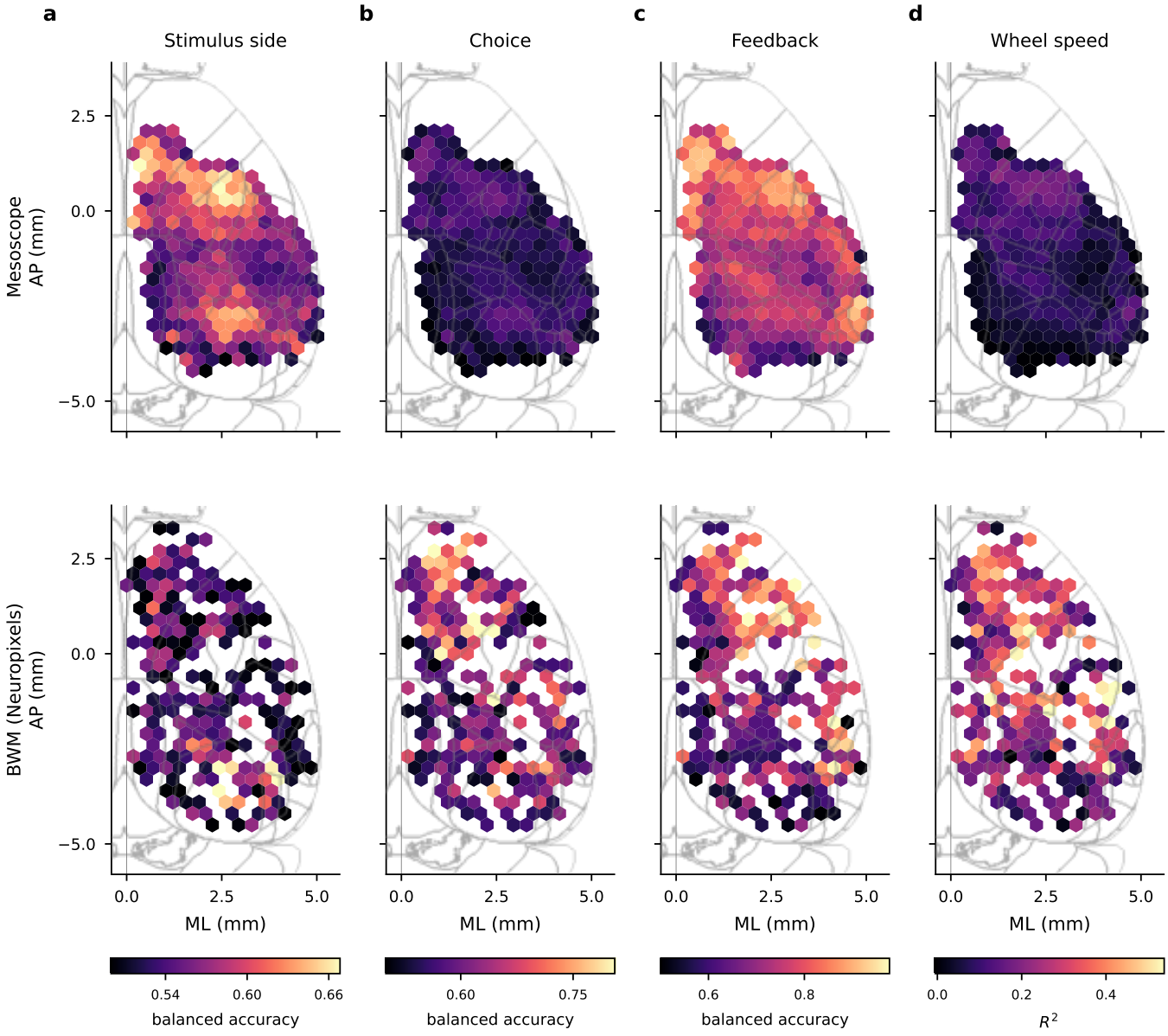


Figure 1: Dorsal cortex maps, mesoscope two-photon imaging (top row; deconvolved activity of isocortex neurons, canonical sessions) vs Brain Wide Map Neuropixels recordings (bottom row; good isocortex units, 2023 release). Each neuron is assigned to a hexagonal bin (0.4 mm vertex to vertex) by its medio-lateral (ML) and antero-posterior (AP) Allen CCF position relative to bregma; both hemispheres are folded onto the right one (ML $\rightarrow$ |ML|). Grey lines: Allen top-view region boundaries. Trials: choice made and reaction time 0.08–2 s (BWM criteria). Colour scales are shared between the two rows. Decoding per session and bin, for every session with $\geq$50 trials and $\geq$10 neurons in the bin: 5-fold stratified cross-validation; binary variables with L2-regularised logistic regression (balanced class weights, regularisation chosen by inner 3-fold cross-validation), wheel speed with ridge regression (penalty chosen by leave-one-out error, folds split by trial). With all neurons, mesoscope bins hold far more neurons than BWM bins, so scores are not comparable across rows; see the 10-neuron version. **a** Stimulus side (left vs right, including 0% contrast) decoded from all neurons of the session in the bin: cross-validated balanced accuracy (chance 0.5), averaged across sessions with $\geq$10 neurons in the bin; activity in the window $[0, 0.1]$ s (BWM) or $[0, 0.3]$ s (mesoscope) relative to stimulus onset. **b** Same for choice (left vs right wheel turn), window $[-0.1, 0]$ s (BWM) or $[-0.3, 0]$ s (mesoscope) relative to movement onset. **c** Same for feedback (reward vs error), window $[0, 0.2]$ s (BWM) or $[0, 0.3]$ s (mesoscope) relative to feedback. **d** Wheel speed (mean absolute wheel velocity) in 0.2 s bins from -0.2 to 1 s around movement onset, decoded from the activity in the same bins: cross-validated R^2 of ridge regression (chance 0).